

Lab 5: Gaussian Blur

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A 7×7 Gaussian filter with coefficient sum 1003 is applied to grayscale pixels. The global version reads the filter from device memory; the shared version cooperatively loads its 49 coefficients and synchronizes before convolution. The three-pixel border is preserved. A CPU reference validates both versions. The graph compares speedup for different 2D blocks. Shared memory has loading and synchronization costs, so its benefit must be measured.

Kernel timing uses `time.time()` after JIT warm-up and CUDA synchronization, averaged over three GPU runs. Device allocation and host/device copies are excluded. CPU measurements are single runs. Speedup is relative to the stated CPU implementation. Actual measurements appear after running the notebook on Colab; no timings are assumed here.